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TRAUMA AND MORTALITY SALIENCE INCREASE COVID FEAR AND DEFENSIVE STRATEGIES

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Abstract – We conducted an experiment to test the effect of mortality salience on in-group bias and COVID fear in a non-clinical population of college students with differing prior exposure to trauma. Our results revealed trauma exposure combined with reminders of mortality increased political in-group favoritism. These findings are consistent with defensive reactions proposed by Terror Management Theory and commonly observed in the general population. However, these same participants reported the highest levels of COVID-19 fear and anxiety, suggesting a deficit in their defensive strategy. Such emotional responses are consistent with reactions observed in persons suffering from PTSD and may reflect trauma-induced deficits in their anxiety buffering system. Our findings suggest that reminders about the fragility of life may be especially painful for those with a history of trauma; however, the nature of reactions to such reminders seems to depend on the intensity and impact of the trauma experienced. We identified a group of adults whose reaction to reminders of death is distinctive. While they exhibit some emotional reactions common among trauma survivors with severe PTSD symptoms, they simultaneously enact behavior not typical of PTSD populations, but rather characteristic of the general population. An awareness of this previously unstudied group will not only inform future research but may provide helpful insight that can be applied to anticipate and mitigate circumstances that set the stage for anxiety, defensiveness, and intergroup conflict.

Keywords: trauma, terror management theory, COVID fear

The recent COVID-19 pandemic created unprecedented health threats and social disruption that dramatically impacted the day-to-day life of people across the globe (Pakpour & Griffiths, 2020). The large-scale social isolation that resulted from masking and quarantine protocols has been linked to negative mental health outcomes for the general population (Kauhanen et al., 2023; Pennix et al., 2022; Satici et al., 2020). There also is reason to believe that the unremitting reminders of death which accompany epidemics such as the COVID-19 pandemic (Firestone, 2020; Menzies & Menzies, 2020) have noteworthy effects on the mental state, behavior, and social functioning of individuals. Terror Management Theory (TMT; e.g., Greenberg et al., 1986; Solomon et al., 1991) provides a framework to explain how individuals react when reminded of their own mortality and how they protect themselves from the anxiety that accompanies concerns about death. According to this theory, people have an inherent need for their existence to have significance and purpose, but the realization of the inevitability of their own mortality causes fear and discomfort. When death is made salient (e.g., when mortality salience is activated), anxiety about one's own mortality is increased and the need to reduce this fear becomes pressing. Individuals are then prompted to engage in a variety of defensive processes. For example, some people consciously deny their physical vulnerability when confronted with threats concerning death. Pyszcynski et al. (1999) found support for this idea by demonstrating that reminders of personal mortality elicited increased thoughts about one's excellent personal health and family longevity.

These findings may help to explain some individuals' behavior in response to reminders of mortality activated by the COVID-19 pandemic (Özgüç et al., 2021; Rupprecht et al., 2022). The pervasive

resistance to masking and distancing protocols observed in some individuals during the pandemic may reflect their defensive belief in their own hardiness and refusal to acknowledge their own risk of contracting the virus.[1]

TMT also argues that when faced with threats about mortality, individuals may cope by engaging in unconscious defensive strategies that serve to bolster their self-esteem and cultural worldview. Specifically, by aligning themselves with cultural belief systems, individuals derive a sense of meaning and significance to their lives, providing self-esteem and protection from mortality-related anxiety. TMT suggests that without this protective anxiety-buffering system, individuals feel vulnerable and overwhelmed when confronted with reminders of death (e.g., Greenberg et al., 1986). Thus, people are motivated to preserve this system of defense and tend to respond favorably to others who reinforce their cultural worldview and unfavorably to those who violate it. A robust body of empirical studies have confirmed that reminders of mortality elicit in-group bias reflected in more favorable attitudes toward members of in-groups (e.g., Arndt et al., 2002; Castano et al., 2011) and decidedly unfavorable attitudes toward outgroup members (Arndt et al., 2002; Gailliot et al., 2007). These biases are consistent with the polarized attitudes that characterize the current U.S. political climate, evidenced by the emotionally charged political disputes often observed between Democrats and Republicans. This persistent political discord may have been exacerbated by fears invoked by the COVID-19 pandemic.[2]

Although TMT provides a framework to understand defensive processes evoked in the general population in response to mortality salience, individuals who have experienced a history of significant trauma seem to respond differently. Research findings suggest that people who report severe PTSD symptoms fail to enact the typical responses predicted by TMT when confronted with reminders of mortality, responding instead with increased anxiety (Abdollahi et al., 2011; Vail et al., 2019). Researchers exploring connections between PTSD and responses to reminders of mortality argue that Anxiety Buffer Disruption Theory (ABDT) provides a useful framework for understanding the atypical reactions commonly observed in individuals with severe PTSD. According to ABDT, traumatic event exposure disrupts the typical anxiety-buffering system invoked by individuals when confronted with reminders of death (Pyszcynski & Kesebir, 2011). This disruption occurs because trauma violates their assumptions about the safety and benevolence of the world, thereby decreasing perceptions of meaning in life and harming self-esteem. (e.g., Herman, 1997; Horowitz, 1976; Janoff-Bulman, 1992; Park & Folkman, 1997). If one's worldview is undermined to the point that it fails to provide sufficient protection, the individual may be left vulnerable and terrorized in the face of mortality salience.

Thus, previous research suggests that TMT may be a framework to understand the general population's reaction to serious public health threats, such as the COVID-19 pandemic. However, TMT does not seem to accurately predict the reactions of individuals with a history of severe trauma leading to a PTSD diagnosis. The question remains as to how individuals with exposure to stress or trauma that does not rise to the level of PTSD might respond to mortality threats in the context of the COVID-19 pandemic. This non-clinical group may experience greater sensitivity to fear-related stimuli, compared to persons without trauma exposure, as a result of their prior experiences and the negative aftermath. They may therefore report greater death-related anxiety and fear associated with COVID-19. This is consistent with ABDT and would provide evidence of trauma-induced deficits in their anxiety-buffering system. However, because these individuals have experienced some trauma, but not enough to rise to the level of PTSD, their anxiety-buffering system is unlikely to have been undermined to the point of collapse. Thus, they may exhibit intensified defensive responses proposed by TMT when reminded of death (even more so than the general population due to their heightened anxiety). Specifically, they may more vigorously cling to their worldview in an attempt to re-establish a sense of security when confronted with the inevitability of death. This, in turn, may lead to increased positive in-group bias for individuals with a prior history of trauma when confronted with threats about death.

We examined these possibilities by manipulating mortality salience in a non-clinical population of college

[1]Although we have a particular interest in examining defensive behavior in relation to COVID anxiety, there are alterative possible interpretations of this resistant behavior (e.g., some individuals may not have perceived the threat to be as strong as what was publicized in the media).

[2]Although in this context in-group bias may have problematic implications, it should be noted that in-group favoritism is adaptive in some contexts and may have originated in part, from evolutionary survival pressures.

students with differing prior exposure to trauma and stress. We examined political in-group bias as well as reported anxiety and fear about death in relation to the COVID-19 virus. We chose this operationalization of death-related anxiety due to the salient, life-threatening reality of COVID-19 at the time of data collection. We tested the following hypotheses:

1) Consistent with TMT theory, the highest levels of positive in-group bias would be reported by participants with a history of relatively high trauma exposure, when reminded of mortality.

2) Consistent with ABDT, the highest levels of fear about COVID-19 would be reported by participants with a history of relatively high trauma exposure, when reminded of mortality.

3) Consistent with TMT, no significant differences in positive in-group bias would occur as a function of prior trauma exposure when mortality salience was not heightened.

4) Consistent with TMT, no significant differences in fear about COVID-19 would occur as a function of prior trauma exposure when mortality salience was not heightened.

5) In line with the application of TMT to the general population, individuals both high and low in reported stress, would report heightened positive in-group bias in the mortality salience condition but not in the control condition.

6) In line with the application of TMT to the general population, individuals both high and low in reported stress, would report lowered fear about COVID-19 in the mortality salience condition but not in the control condition.

Method

Participants

Data was collected during the Fall of 2020, during a highly stressful time period of the COVID-19 pandemic. The University Institutional Review Board deemed the project to be exempt from oversight. Federal guidelines for the ethical treatment of human participants were followed. Participants were recruited through college classrooms and tested in group settings. No participants self-identified as having a diagnosis of PTSD. Participants were 19 men and 54 women with a mean age of 19.86 (SD = 2.28). Participants self-reported their racial identity as Caucasian (n = 34), African American (n = 26), Hispanic (n = 10), and other (n = 3). Eighty percent identified as heterosexual, while the remainder identified as non-heterosexual. Participants received extra credit for taking part in the study.

Materials

Participants read an informed consent document and after agreeing to take part, began by responding to the 21-item University Stress Scale (Stallman, 2008). This scale measures the number of stressful life events an individual had experienced in the past year, specifically focused on typical stressors experienced by young adults. Examples were failing an important class, experiencing a serious illness or injury, and losing someone close to them. Participants indicated their exposure to each stressor on a 4-point scale where 1 represented “not at all” and 4 represented “constantly.” Responses were summed (with possible sums ranging from 21 to 84) and then averaged to create a composite stress score for each participant. Cronbach’s α for this measure was .79, suggesting acceptable internal reliability.

Next, participants responded to the 18-item Trauma and Life Events Checklist (Carr et al., 2018). This scale measures the extent to which individuals experienced traumatic events at some time in their past. Events included exposure to war, bullying or harassment at school, and witnessing physical violence in the home. Participants indicated whether they had never experienced, only witnessed, or personally experienced each event. Participants earned two points for each event experienced and one point for each event witnessed. These points were summed (with possible sums ranging from zero to 40) and then averaged to achieve a trauma score for each participant, with higher scores indicating greater exposure to trauma. Cronbach’s α for this measure was .88, suggesting acceptable internal reliability.

Afterward, participants were randomly assigned to either write about what would happen to them if they died (mortality salience condition) or experienced physical pain (control condition; based on Rosenblatt et al., 1989). Next, participants responded to the 20-item Positive and Negative Affect Scale (Watson et al., 1988) whereby they indicated the extent to which they were feeling a series of 10 positive and 10 negative emotions, such as enthusiastic, excited, irritable, and anxious. This scale is the most commonly used task in TMT experiments to create a delay between the manipulation and the dependent measures, strategically moving the thoughts of death to a subconscious level while still being highly accessible (Burke et al., 2010). Thus, this scale was used only as a distractor task and not included in data analysis.

Participants were provided brief narratives of two fictitious people created by the researchers. The first narrative described a person who agreed with their political viewpoint, and the second narrative described a

person who disagreed with their political viewpoint. The narrative read "Please imagine a person who agrees (disagrees) with your political stance and plans to vote the way you do (opposite of you) in the upcoming election." After each narrative, participants responded to 13 descriptors derived from Byrne (1971) and Greenberg et al. (1990). Participants used a 5-point scale, ranging from 1 "very slightly or not at all" to 5 "extremely" to indicate to what extent each descriptor matched the imagined person. Descriptors included 13 positive attributes such as intelligent, knowledgeable about current events, kind and trustworthy. Ratings of positive descriptors in relation to the individual who matched participants' political views were summed (with possible sums ranging from 13 to 65) and then averaged to create a composite score for each participant that we defined as "positive ingroup bias". Cronbach's α for this measure was .87, suggesting acceptable internal reliability.

Next, participants responded to two sub-scales (Death Acceptance and Thoughts about Death) from the Death Anxiety Scale-R (Thomás-Sábado et al., 2006). We modified the scale such that references to death were changed to refer to COVID-19 anxiety specifically. For example, the item "I think I am more afraid of death than most people" was modified to read "I am more afraid of dying from COVID-19 than most people." Participants responded to nine statements on a 5-point scale where 1 represented "totally disagree" and 5 represented "totally agree." Responses were summed (with possible sums ranging from 9 to 45) and then averaged to create a composite score for each participant that we defined as "COVID fear." Cronbach's α for this measure was .76, suggesting acceptable internal reliability. Participants also responded to four demographic items including sexual orientation, age, race, and gender.

Results

We conducted a median split on the composite trauma scores to create high trauma and low trauma groups. We followed the same procedure to create low stress and high stress groups. The mean score for COVID fear was 2.68 (SD = .76), representing "neutral" on our scale.

We ran a 2(trauma level) X 2(experimental condition) MANOVA to examine positive in-group bias, and COVID fear. Compared to the other three categories, participants with high trauma in the mortality salience condition reported the highest levels of positive in-group bias [$F(3, 70) = 4.06$, $p = .049$] and the highest levels of COVID fear [$F(3, 70) = 6.47$, $p = .013$].

A 2(stress level) X 2(experimental condition) MANOVA on the same dependent variables revealed no significant interactions ($p > .05$ in all cases).

Discussion

We found support for our hypothesis that adults who had experienced high trauma would exhibit the highest levels of in-group bias when reminded of their own mortality. Trauma exposure combined with reminders of mortality increased adults' political in-group favoritism. These findings are consistent with the defense mechanism proposed by TMT that reflects an effort to bolster one's worldview as a means of protection from terror associated with awareness of the inevitability of death. As we hypothesized, these same adults reported the highest levels of COVID fear, suggesting a deficit in their defensive strategy predicted by ABDT. In other words, reminders of mortality elicited some of the defense mechanisms proposed by TMT; however, these strategies failed to successfully decrease the individual's anxiety, as evidenced by their reported heightened fear of COVID. This non-clinical cohort represents a combination of the behaviors seen in the general population and among PTSD sufferers, likely reflecting their status of experiencing trauma levels greater than that of the general population but less than individuals with PTSD.

Participants in the low trauma condition reported similar levels of in-group favoritism and fear of COVID regardless of whether they were reminded of their mortality or not. Similarly, and contradicting our hypothesis, we did not see differences in favoritism or COVID fear across individuals in high and low stress categories whether they were reminded of their mortality or not. One take-home message is that adults who had experienced high levels of trauma functioned differently than others in our sample. This finding suggests that those with a non-clinical history of trauma may possess a unique sensitivity to threatening stimuli, rendering them vulnerable to terror about the possibility of death in a way that simple stress does not. At the same time, this non-clinical high trauma group responded to threatening stimuli in a way that individuals with PTSD do not, as prior research shows those with PTSD fail to invoke defensive reactions in response to mortality salience (e.g., (Abdollahi et al., 2011; Vail et al., 2019). Our findings provide evidence of the influential role that specific levels of trauma might play in TMT.

The question remains as to why morality reminders did not consistently elicit in-group favoritism and anxiety for our low and high stress participants as might be expected according to TMT theory and as we hypothesized. One possible explanation is that our mortality reminder occurred in the context of the COVID-19 pandemic. Social media and daily life experiences may have created a mild but chronic mortality saliency

condition, which was more influential than our short-lived exercise for all of our participants except those who had experienced high levels of prior trauma. Indeed, collective health crises, including the COVID-19 pandemic, increase social media usage and reliance (Freberg et al., 2013; Pérez-Escoda et al., 2020).

In sum, our data suggests that reminders about the fragility of life may be especially painful for those with a history of trauma; however, the extent to which adults react to such reminders seems to depend on the intensity and impact of the trauma experienced. Our study suggests that there may be a group of adults whose reaction to reminders of death is not entirely accounted for by either TMT or ABDT. Rather, they fall in the gap between TMT theory which explains the reactionary behavior of the general population when faced with mortality threats and ABDT theory which explains the reactionary behavior of individuals with PTSD when faced with mortality threats. An awareness of this previously unstudied group will not only inform future research but may provide helpful insight that can be applied to anticipate and mitigate circumstances that set the stage for anxiety, defensiveness, and intergroup conflict. This insight might be especially applicable to young adults, given that research (e.g., Cusack et al., 2019) suggests that exposure to traumatic events, without an accompanying diagnosis of PTSD, is common amongst college students.

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